

ABHILASH G

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SUMMARY

A results-driven Electronics and Communication Engineering undergraduate with a strong foundation in VLSI design, Python, and Cadence Virtuoso. Currently applying hands-on skills in IoT and system integration as an intern at GoodEarth Pvt Ltd, focusing on home automation. Seeking to leverage a robust technical skill set and project-based experience to contribute to innovative engineering challenges, while actively pursuing both personal growth and organizational success.

EDUCATION

Sir M Visvesvaraya Institute of Technology, Bengaluru Bachelor of Engineering in Electronics and Communication Engineering CGPA – 7.58	Dec 2022 – Present
Sri Chaithanya Techno School, Bengaluru Pre - University PCMC (CBSE Percentage – 85.4%)	2020 – 2022
Delhi Public School Bangalore North, Bengaluru 10 th CBSE Percentage - 86.8%	2010 - 2020

WORK EXPERIENCE

Project Intern (Smart Home & IoT Systems): GoodEarth Pvt. Ltd	Aug 2025 – Oct 2025
- Designed and deployed end-to-end smart home automation solutions, integrating lights, fans, sensors, and actuator across multiple platforms for seamless control and remote accessibility. - Implemented IoT-based monitoring systems for energy consumption, irrigation, and water leakage, enabling real-time data visualization, analytics, and alert notifications via Home Assistant. - Developed and configured hardware-software integrations using ESP32 microcontrollers, Raspberry Pi, Aqara ecosystem devices, and sensors, ensuring reliable communication and automation. - Managed system architecture, device provisioning, and automation workflows, delivering scalable, user-friendly, and vendor-agnostic solutions across residential environments.	

SKILLS

- Programming Languages:** C/Embedded C (Basics), Python, Verilog HDL, SystemVerilog, SQL (Basics), yaml
- Tools & Software:** Cadence Virtuoso, MATLAB, LabVIEW, Multisim, Arduino IDE, Xilinx Vivado, Home Assistant

PROJECTS

1. Smart Home Floor Planning & Automation

- Designed and deployed a 57-device automation ecosystem including 34 lights, 13 fans, 2 LED strips, and 8 wall lights.
- Integrated MagicPad S1 Plus (controlling 13 lights, 6 fans, 3 wall lights) and Aqara Z1 Pro switches for the remaining devices via Aqara Hub M3.
- Created a custom digital floor plan with Home Assistant Picture Elements, mapping devices for direct visual control.
- Configured room-level master switches and global scenes (*Goodnight, Good Morning*) for energy efficiency and user convenience.

2. Energy Monitoring System

- Built a live energy monitoring solution for BESCO grid and DG power consumption with cost and usage using the Selec EM4M 4PC 100a meter breakdown.
- Deployed Raspberry Pi running Home Assistant to process and visualize energy data.
- Configured automation rules for load balancing and efficient energy usage.

3. Smart Irrigation System

- Implemented an automated irrigation system using Yardian Pro, solenoid valves, and soil moisture sensors.
- Integrated with Home Assistant for condition-based watering schedules.
- Reduced water wastage through real-time monitoring and automation logic.

4. Water Leak Detection System

- Designed an IoT-based leak detection system using ESP32-Nano, YF-S201 flow sensor, and Raspberry Pi.
- Configured ESPHome for real-time flow measurement, usage logging, and abnormal flow alerts.
- Automated push notifications via Home Assistant mobile app to prevent damage.

5. Color Sorting Machine using ESP32

- Designed and implemented an automated color-sorting system using ESP32, TCS3200 color sensor, and 3 servo motors.
- Programmed sorting logic in Arduino IDE, leveraging I2C communication and PWM control to detect RGB-colored objects and direct them into designated bins.

CERTIFICATIONS & WORKSHOPS

ESP32 Robotics Workshop (NEWRR0)

Nov 2024

- Built a wheeled robot with object-following, obstacle avoidance, and Bluetooth control features.
- Programmed sensors and motors using Arduino IDE, interfacing hardware with the ESP32 microcontroller for autonomous and remote operation.

NIELT Online Certification – VLSI Design

Sep 2024

- Completed a course covering CMOS logic design, ASIC & FPGA architecture, and digital logic families.
- Conducted hands-on simulations using Xilinx Vivado to design and test digital circuits.